

LEDGER · 2026-09-04 · MATERIALS SCIENCE / PHOTOPOLYMER CHEMISTRY

Ultra-Low Stress Thiol-Ene Photopolymer Conformal Coatings (USTE-CC) via Step-Growth Radical Photopolymerization

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	HumiSeal UV40
Entry application	high-reliability aerospace and defense printed circuit boards (PCBs)
Measured advantage	>2.1x reduction in polymerization shrinkage stress (1.0-1.2 MPa vs 2.5-2.6 MPa for HumiSeal UV40)
Time to first revenue	18 months
Gross margin at parity	88.2%
Startup CapEx	\$125,500
Payback from first sale	0.1 months

Capital productivity

199.01x

Regulatory risk

Moderate — qualification costs reported (\$85,000 estimated)

Why it failed

- Voided by: Absence search missing
- Absence asserted, search unreported

Assessment

The concept demonstrates strong scientific backing and a >2.1x reduction in shrinkage stress against the market leader HumiSeal UV40. However, it fails because it lacks a structured Market Absence Ledger section with numeric search metrics and named finds. Additionally, the report acknowledges significant commercial barriers regarding shadow cure deficiencies and aerospace qualification timelines.

Also flagged

- Duplicate references inflating the evidence base
- Price parity asserted, not sourced
- runsheet_no_quantities

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-09-04 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

biological or regulated processes that require appropriate expertise, facilities, permits and safety controls. Independently verify every figure, citation and regulatory requirement before committing capital.